

Data Mining: Concepts and Techniques

Chapter 1 – Introduction

Based on Data Mining: Concepts
& Techniques (4th Edition)
Prepared for academic teaching

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Chapter Roadmap

What is Data Mining? -	1.The necessity and motivation
Knowledge Discovery Process -	1.The KDD framework
Data Diversity -	1.Types of data we can mine
Knowledge Variety -	1.Different patterns and models
Interdisciplinary Nature -	1.Confluence of multiple fields
Real-World Applications -	1.Where data mining shines
Societal Impact -	1.Ethics, privacy, and future

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What is Data Mining?

- We live in the era of massive data generation
- Data mining discovers hidden patterns and knowledge from data
- Transforms raw data into useful information



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Why Data Mining is Needed

We are drowning in data but starving for knowledge

The Explosive Growth:

Terabytes → Petabytes → Zettabytes daily

Sources: Business, science, social media, IoT, healthcare

The Critical Gap:

Data storage capacity has far outpaced our **manual analysis capabilities**

Traditional methods fail with massive, complex datasets

The Solution:

Data Mining: Automated discovery of valuable patterns

Transforms raw data into **actionable intelligence**

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Definition of Data Mining

Formal Definition: "The process of discovering interesting, non-trivial, implicit, previously unknown, and potentially useful patterns or knowledge from large amounts of data."

The "Mining" Analogy:

- Gold mining: Extract precious nuggets from tons of rock
- Data mining: Extract knowledge nuggets from terabytes of data
- Technically should be called "knowledge mining from data"

Related Terms:

- Knowledge Discovery in Databases (KDD)
- Pattern discovery, knowledge extraction
- Data archaeology, information harvesting
- **Modern term: Data Science** (broader context)

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Example: Data Mining in Search Engines

Billions of queries generated daily

Patterns emerge only after aggregation

Example: Google Flu Trends

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Data Mining and Knowledge Discovery (KDD)

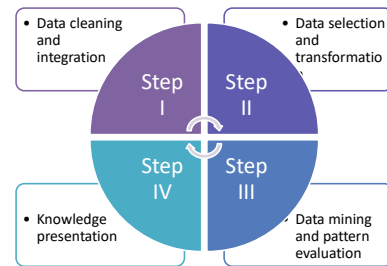
Data mining is a key step in KDD

KDD is an iterative, multi-step process

Includes preprocessing and evaluation

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Steps in the KDD Process



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Data Mining as One Step in KDD

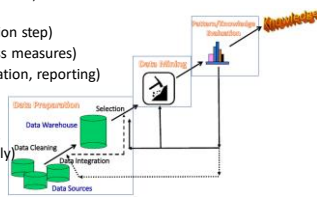
- **Data Preparation** (cleaning, integration, transformation, selection)
- **Data Mining** (core pattern extraction step)
- **Pattern Evaluation** (interestingness measures)
- **Knowledge Presentation** (visualization, reporting)

Critical Insight:

- **Data mining ≠ entire KDD process** (though often used interchangeably)

- **80% effort in data preparation** - "garbage in, garbage out"

- Iterative process with human-in-the-loop feedback



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Types of Data for Mining

- Structured data
- Semi-structured data
- Unstructured data



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Structured vs Unstructured Data



Structured:

Relational databases, data cubes, CSV files
Clear schema, fixed attributes

Example: Sales transactions with product ID, price, timestamp



Semi-structured:

XML, JSON, graphs, sequences
Some structure but flexible

Example: Social network data (users, connections, posts)



Unstructured:

Text documents, images, audio, video

No predefined model
Example: Medical records, surveillance footage

Stored Data vs Streaming Data



- **Stored:** finite datasets
- **Streaming:** continuous, infinite flows
- Streaming requires real-time mining

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1.4 Mining various kinds of knowledge Data Mining Tasks

Descriptive data mining

- Descriptive methods summarize data

Predictive data mining

- predictive methods forecast unknown values.

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Knowledge Variety - What Patterns Can We Find?

Two Broad Categories:

1. Descriptive Mining

Characterization: Summarize data characteristics

Frequent Patterns: Itemsets, sequences, structures

Clustering: Group similar objects

Association Rules: "If A then B" relationships

2. Predictive Mining

Classification: Predict categorical labels

Regression: Predict numerical values

Deep Learning: Automatic feature learning

Outlier Detection: Find anomalies

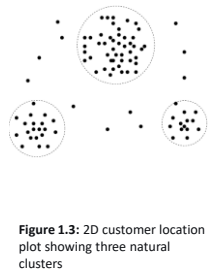


Figure 1.3: 2D customer location plot showing three natural clusters

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Multidimensional Data Summarization

- Summarizes data across dimensions
- Uses data cubes and OLAP
- Supports roll-up and drill-down

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Frequent Pattern Mining

- Frequent itemsets
- Sequential patterns
- Structured patterns

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Association Rule Mining - A Closer Look

- **Market Basket Analysis:**
- buys(X, "computer") $\Rightarrow$ buys(X, "webcam") [support=1%, confidence=50%]
- **Key Metrics:**
- **Support:** Frequency of pattern in data (1% of transactions)
- **Confidence:** Conditional probability (50% of computer buyers also buy webcam)
- **Types of Rules:**
- **Single-dimensional:** computer $\Rightarrow$ webcam
- **Multidimensional:** age=20-29 $\wedge$ income=\$40-49K $\Rightarrow$ buys(laptop)
- **The Challenge:**
- Generate millions of possible rules
- Need thresholds for minimum support & confidence
- Interestingness measures filter trivial/obvious rules

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Classification and Regression

Classification: Predict categorical class labels

Model forms: Decision trees, rules, neural networks, SVM
Example: Loan approval (approve/deny), disease diagnosis

Regression: Predict numerical values

Model forms: Linear regression, regression trees

Example: House price prediction, sales forecasting

Process:

Training: Learn model from labeled examples

Testing: Evaluate on unseen data

Application: Predict for new instances

Figure 1.4: Different classification model representations:

(a) IF-THEN rules, (b) decision tree, (c) neural network

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Clustering & Outlier Detection

Clustering: Grouping Without Labels

Goal: Maximize intra-cluster similarity, minimize inter-cluster similarity

Applications: Market segmentation, social network communities

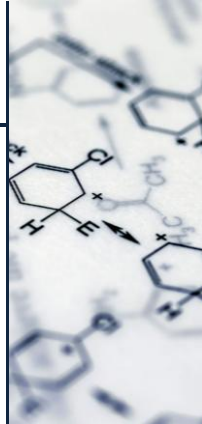
Methods: K-means, hierarchical, density-based

Outlier Detection: Finding the Exceptions

Anomalies: Fraud, defects, network intrusions

Approaches: Statistical, distance-based, density-based

Key insight: "One person's noise is another person's signal"



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Deep Learning - The Game Changer

Automatic Feature Engineering

Traditional ML: Manual feature extraction (tedious, task-specific)

Deep Learning: Learns hierarchical representations automatically

Neural Network Architectures:

Feedforward: Standard MLP for tabular data

Convolutional (CNN): Images, spatial data

Recurrent (RNN): Sequences, time-series, text

Graph Neural Networks: Social networks, molecules

Why Now?

Big data + GPU computing + algorithmic advance
Breakthroughs in image recognition, machine translation, game playing

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Outlier Analysis

Detects rare or abnormal patterns

Important for fraud detection

Uses statistical and distance methods

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Interestingness of Patterns

Objective Measures:

- **Support & Confidence:** For association rules
- **Accuracy, Precision, Recall:** For classification
- **Silhouette Coefficient:** For clustering quality
- **Statistical significance:** p-values, hypothesis testing

Subjective Measures:

- **Unexpectedness:** Contradicts prior beliefs
- **Actionability:** Can act on the knowledge
- **Novelty:** New insight not just common sense
- **Interpretability:** Humans can understand it

The Completeness Dilemma:

- Can we generate **all** interesting patterns? (No, too many)
- Can we generate **only** interesting patterns? (Goal with constraints)

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Data Mining - A Confluence of Disciplines

The Interdisciplinary Tree:

Roots (Core Disciplines):

- **Statistics:** Inference, testing, distributions
- **Machine Learning:** Algorithms that learn from data
- **Database Systems:** Scalable storage and querying
- **Information Retrieval:** Text, search, relevance

Branches (Supporting Fields):

- Pattern recognition, visualization, algorithms
- High-performance computing, distributed systems
- Domain knowledge (biology, business, social science)
- **Figure 1.5:** Venn diagram showing data mining at intersection of multiple fields

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Statistics & Machine Learning Connection

Statistics in Data Mining:

- **Descriptive stats:** Summarize data (mean, variance)
- **Inferential stats:** Hypothesis testing, confidence intervals
- **Statistical models:** Regression, Bayesian networks
- **Validation:** Statistical significance of patterns

Machine Learning Synergy & Differences:

- **Shared problems:** Classification, clustering
- **Data mining focus:** Scalability on massive datasets
- **Weak supervision:** Limited labels (semi-supervised, active learning)
- **Beyond ML tasks:** Frequent patterns, network mining

Figure: Semi-supervised learning with few labeled + many unlabeled examples

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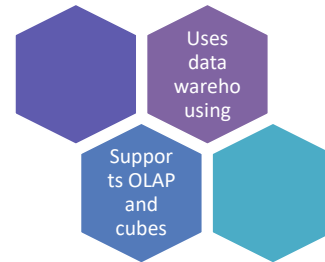
Data Mining & Machine Learning

Supervised vs
unsupervised
learning

Scalability and weak
supervision

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Data Mining & Databases



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Database Technology & Data Science

Database Foundations:

- **Data warehousing:** Integrated, historical data
- **OLAP & Data Cubes:** Multidimensional analysis
- **Query optimization:** Efficient data access
- **Indexing:** Fast search and retrieval

Data Science: The Fourth Paradigm

- **Jim Gray's vision:** Empirical → Theoretical → Computational → Data-Driven
- **Data Science unifies:** Statistics + ML + Data Mining + Domain Knowledge
- **Curriculum:** Computation, statistics, visualization, ethics, storytelling

The Big Data Era:

- **Volume, Velocity, Variety, Veracity, Value (5 V's)**
- **Scalability challenges:** Distributed algorithms, streaming

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Applications of Data Mining

1. Business Intelligence

- Customer segmentation, churn prediction
- Market basket analysis, recommendation systems
- **Example:** Amazon's "customers who bought this also bought..."

2. Web Search Engines

- Query classification, ranking algorithms
- Personalized search, query suggestion
- **Challenge:** Real-time processing of billions of queries

3. Social Media & Networks

- Community detection, influence analysis
- Sentiment analysis, trend detection
- **Example:** Facebook friend recommendations, Twitter trend mining

4. Biomedical & Healthcare

- Genomic sequence mining, drug discovery
- Medical image analysis, electronic health records
- **Example:** IBM Watson for oncology, pandemic prediction models

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Business Intelligence



Customer analysis



Market prediction



Decision support

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Web Search Engines



MASSIVE
DISTRIBUTED DATA



REAL-TIME MINING



QUERY
UNDERSTANDING

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Social Media & Networks



COMMUNITY
DETECTION



SENTIMENT
ANALYSIS



INFORMATION
DIFFUSION

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Healthcare & Biology



GENOMICS AND
PROTEOMICS



MEDICAL
IMAGING



ELECTRONIC
HEALTH RECORDS

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Societal Impact & Ethical Considerations

The Dual-Edged Sword:

- **Benefits:** Scientific discovery, business efficiency, personalized services
- **Risks:** Privacy invasion, discrimination, surveillance

Key Concerns:

- **Privacy:** Re-identification from "anonymized" data
- **Bias:** Algorithms perpetuating societal biases
- **Transparency:** "Black box" models making life-altering decisions

Emerging Solutions:

- **Privacy-preserving data mining** (differential privacy)
- **Fairness-aware algorithms** (bias detection and mitigation)
- **Explainable AI** (interpretable models)
- **Regulation:** GDPR, CCPA, AI ethics guidelines

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Chapter Summary

Key Takeaways:

1. **Data mining** extracts knowledge from massive data—essential in our data age
2. **KDD process** includes preparation, mining, evaluation, presentation
3. **Diverse data types** require different mining techniques
4. **Multiple pattern types:** Descriptive (what happened) & Predictive (what will)
5. **Interdisciplinary:** Draws from statistics, ML, databases, domain knowledge
6. **Applications everywhere:** Business, science, healthcare, social media
7. **Societal impact** requires ethical consideration and responsible use

Looking Ahead:

- **Big data challenges:** Volume, velocity, variety
- **Interpretability vs. accuracy** trade-off
- **Automated machine learning (AutoML)**
- **Ethical frameworks** for responsible AI

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Key Terms & Concepts

- Data Mining / Knowledge Discovery
- Structured vs. Unstructured Data
- Descriptive vs. Predictive Mining
- Association Rules (support, confidence)
- Classification & Regression
- Clustering & Outlier Detection
- Deep Learning & Feature Engineering
- Pattern Interestingness (objective/subjective)
- KDD Process (cleaning → mining → evaluation)
- Big Data, Data Science, Business Intelligence

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References & Further Reading

Primary Source:

- Han, J., Kamber, M., & Pei, J. (2023). *Data Mining: Concepts and Techniques* (4th ed.). Morgan Kaufmann.

Supplementary:

- Witten, I. H., et al. (2016). *Data Mining: Practical Machine Learning Tools*
- Provost, F., & Fawcett, T. (2013). *Data Science for Business*
- **Online Courses:** Coursera's Data Mining specialization, edX Analytics courses

Interesting Articles:

- "The Unreasonable Effectiveness of Data" (Halevy, Norvig, Pereira)
- "Critical Questions for Big Data" (boyd & Crawford)
- "Weapons of Math Destruction" (O'Neil) - on algorithmic bias

Hands-On Practice:

- **Tools:** Weka, RapidMiner, scikit-learn, TensorFlow
- **Datasets:** UCI Machine Learning Repository, Kaggle competitions

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Q&A / Discussion Points

	What data mining application most surprised you or seems most transformative?
	Where have you personally encountered data mining results (recommendations, fraud alerts, etc.)?
	How should we balance data mining benefits against privacy risks?
	Can you think of a novel application for data mining in your field of interest?
	What ethical guidelines should data mining practitioners follow?

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Thanks

Data is the new oil/gold

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